RAJALAKSHMI ENGINEERING COLLEGE

Sanjay Krishna R 251801290-2025-AIDS S2

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|---|---|
| **Started on** | Saturday, 12 September 2026, 1:48 PM |
| **State** | Finished |
| **Completed on** | Saturday, 12 September 2026, 1:49 PM |
| **Time taken** | 35 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories.

If he has eaten $i$ burgers with c calories each, then he has to run at least $3^i * c$ kilometers to burn out the calories. For example, if he ate 3

burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3^1 * 3) + (3^2 * 2) = 1 + 9 + 18 = 28$.

But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance

he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach to solve the problem.

**Input Format**

First Line contains the number of burgers

Second line contains calories of each burger which is n space-separate integers

**Output Format**

Print: Minimum number of kilometers needed to run to burn out the calories

**Sample Input**

3

5 10 7

**Sample Output**

76

**For example:**

| Test | Input | Result |
|---|---|---|
| Test Case 1 | 3<br>1 3 2 | 18 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>
int compare(const void *a, const void *b) {
    return (*(int *)b - *(int *)a); // Descending order
}
int main() {
    int n;
    scanf("%d", &n);
    int calories[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &calories[i]);
    }
    qsort(calories, n, sizeof(int), compare);
    long long total = 0;
    long long power = 1; // 3^0
    for (int i = 0; i < n; i++) {
        total += calories[i] * power;
        power *= n;
    }
    printf("%lld\n", total);
    return 0;
```

```
22 }
```

| | Test | Input | Expected | Got | |
|---|---|---|---|---|---|
| ✔ | Test Case 1 | 3<br>1 3 2 | 18 | 18 | ✔ |
| ✔ | Test Case 2 | 4<br>7 4 9 6 | 389 | 389 | ✔ |
| ✔ | Test Case 3 | 3<br>5 10 7 | 76 | 76 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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